

Individual differences in white matter anatomy predict dissociable components of reading skill in adults[☆]

Suzanne E. Welcome^{a,*}, Marc F. Joanisse^b

^a Department of Psychology, University of Missouri-St. Louis, St. Louis, MO, USA

^b Department of Psychology and Brain and Mind Institute, The University of Western Ontario, London, ON, Canada



ARTICLE INFO

Article history:

Accepted 25 March 2014

Available online 2 April 2014

Keywords:

DTI

Phonological

Primary auditory cortex

Inferior fronto-occipital fasciculus

Uncinate fasciculus

ABSTRACT

We used diffusion tensor imaging (DTI) to investigate relationships between white matter anatomy and different reading subskills in typical-reading adults. A series of analytic approaches revealed that phonological decoding ability is associated with anatomical markers that do not relate to other reading-related cognitive abilities. Thus, individual differences in phonological decoding might relate to connectivity between a network of cortical regions, while skills like sight word reading might rely less strongly on integration across regions. Specifically, manually-drawn ROIs and probabilistic tractography revealed an association between the volume and integrity of white matter underlying primary auditory cortex and nonword reading ability. In a related finding, more extensive cross-hemispheric connections through the isthmus of the corpus callosum predicted better phonological decoding. Atlas-based white matter ROIs demonstrated that relationships with nonword reading were strongest in the inferior fronto-occipital fasciculus and uncinate fasciculus that connect occipital and anterior temporal cortex with inferior frontal cortex. In contrast, tract volume underlying the left angular gyrus was related to nonverbal IQ. Finally, connectivity underlying functional ROIs that are differentially active during phonological and semantic processing predicted nonword reading and reading comprehension, respectively. Together, these results provide important insights into how white matter anatomy may relate to both typical reading subskills, and perhaps a roadmap for understanding neural connectivity in individuals with reading impairments.

© 2014 Elsevier Inc. All rights reserved.

Introduction

Reading is a complex skill that requires integration of multiple cognitive abilities and processes, including visual perception, letter-sound knowledge, word meaning, and attention. Not surprisingly, neuropsychology has identified multiple brain regions subserving reading, presumably due to the different types of knowledge represented. Most recently, evidence has come from fMRI studies that have sought to identify how different cortical regions are associated with different subcomponents of reading processes. We and others have recently identified ways in which the engagement of cortical regions subserving reading can vary in the normal population of readers, and in a way that reflects individual differences in reading subskills (Clements-Stephens et al., 2012; Welcome and Joanisse, 2012). There is increasing interest in the gray matter sources of individual differences in reading (Welcome et al., 2011) as they permit us to better assess the contribution of specific

neural subregions to specific cognitive subskills. However, questions remain about how individual differences in reading are linked to white matter tracts responsible for the coordination of these many cortical regions during reading.

It has long been hypothesized that efficient language processing relies on white matter pathways to connect distant cortical regions (Wernicke, 1874). This hypothesis has been supported by lesion data demonstrating that disconnections of language-relevant cortex can produce reading deficits even if the cortical regions themselves are spared (Dejerine, 1892). In the present study, we focus on reading ability in humans, an ability that critically relies on the ability to coordinate multiple brain regions supporting different sub-processes (Sandak et al., 2012). We have taken advantage of recent advances in diffusion tensor imaging (DTI) analyses to explore the hypothesis that individual differences in white matter pathway volume and integrity can predict component reading skills in typical-reading adults. DTI-weighted MRI scans permit the characterization of water diffusivity in the brain, and are thus well suited to characterizing the integrity of white matter tracts at a scale of 1–3 mm. We explored relationships between skill in different aspects of reading and white matter tracts identified using DTI in order to tease apart the roles of various white matter pathways in reading.

[☆] This research was supported by A Discovery grant and an Accelerator Award from the Natural Sciences and Engineering Research Council of Canada.

* Corresponding author at: Department of Psychology, University of Missouri-St. Louis, One University Boulevard, 325 Stadler Hall, St. Louis, MO 63121-4499, USA. Fax: +1 314 516 5393.

E-mail address: welcomes@umsl.edu (S.E. Welcome).

Various white matter pathways have been associated with language both because of their anatomy and as consequences of lesions. While the focus of some studies has been spoken language, and others have focused on reading, there is reason to believe that linked mechanisms are engaged by both spoken language and written language. Specifically, both require phonological and semantic processing, two processes which appear to be subserved by partially distinct cortical networks. The pathway most frequently associated with language is the arcuate fasciculus (AF), which travels between the ventrolateral frontal lobe and superior/middle temporal regions, connecting classical Broca's and Wernicke's areas. It has been hypothesized that this pathway is involved with sensory-motor mapping of spoken language (Saur et al., 2008) and subcortical stimulation along the AF has been shown to result in phonological disturbances (Duffau et al., 2002). Recent DTI work suggests that the AF can be divided into a direct pathway (traversing the whole path) and an indirect pathway in which separate portions travel between frontal and inferior parietal regions and between inferior parietal and temporal regions (Catani et al., 2005). Evidence from aphasic patients suggests that damage to the direct pathway results in difficulty processing phonological aspects of language, while damage to the indirect pathway results in difficulties with semantic aspects (Catani et al., 2005).

In addition to the AF, neuroimaging data suggest that other white matter pathways may also be involved in processing language, particularly word-based semantic memory (Duffau et al., 2005; Saur et al., 2008) and written word forms (Cohen and Dehaene, 2004). In a patient with acquired pure alexia as a result of surgery, the inferior longitudinal fasciculus degenerated while the AF remained intact (Epelbaum et al., 2008). This finding suggests that pathways other than the AF between occipital and temporal cortex are necessary for efficient reading. Anatomical alteration of the uncinate fasciculus, which connects the anterior temporal lobe with ventral frontal regions, has also been associated with difficulties in picture naming (Grossman et al., 2004; Lu et al., 2002). Finally, subcortical stimulation of the inferior fronto-occipital fasciculus, which passes through the temporal lobe, induced semantic paraphasias (Mandonnet et al., 2007).

Further, there is evidence that white matter anatomy is disrupted in children and adults with dyslexia. Adults with poor reading skills showed less fractional anisotropy (FA, an indicator of white matter coherence as measured by DTI) in bilateral temporo-parietal regions than typical readers (Klingberg et al., 2000). In adults with a history of reading and spelling difficulties, FA was reduced in bilateral fronto-temporal regions as well as a left temporo-parietal region (Steinbrink et al., 2008). When FA was examined within specific white matter ROIs, the posterior superior longitudinal fasciculus showed decreased FA in children with dyslexia (Carter et al., 2009), while FA within the arcuate fasciculus was negatively correlated with phonological awareness (Yeatman et al., 2011). A tract-based spatial statistics approach revealed broader reductions in bilateral white matter tracts in frontal, temporal, parietal, and occipital regions in adults with dyslexia (Richards et al., 2008).

One question that arises from prior findings of alterations in white matter anatomy in individuals with dyslexia is whether these differences in white matter anisotropy and/or coherence reflect the tail end of a normal distribution throughout the population, or are a marker of frankly atypical neural organization in dyslexia. Such a question motivates our current interest in individual differences within the population of normal-range readers. In this view, the language disruptions that come about as a result of lesions represent an extreme version of brain/behavior relationships that exist across the spectrum of language skill. Prior work supports the view that connectivity differences associated with dyslexia are mirrored by relationships between individual differences in reading and white matter microstructure. Word reading ability was positively correlated with FA in left temporo-parietal white matter in children (Beaulieu et al., 2005) and adults (Klingberg et al., 2000). In the ROI identified in adults by Klingberg et al., children

showed relationships between FA and coherence and standardized measures of word reading, nonword reading, spelling, and rapid naming (Deutsch et al., 2005). Children with dyslexia whose reading improved over time showed greater FA in the right superior longitudinal fasciculus, suggesting that greater integrity of this right hemisphere pathway supported reading improvement (Hoeft et al., 2011). Similarly, FA in the left anterior centrum semiovale increased with improvement in phonological decoding as a result of reading remediation in 8–10 year old poor readers (Keller and Just, 2009).

While there appears to be consensus that left temporo-parietal white matter organization is related to general reading skill, the precise nature of this relationship remains unclear. While the majority of fibers in the temporo-parietal region identified by Klingberg et al. (2000) had an anterior–posterior orientation, and could thus be part of the arcuate fasciculus, other studies have noted that the majority of fibers in regions that show associations with reading have an inferior–superior orientation, consistent with the corona radiata (Beaulieu et al., 2005; Deutsch et al., 2005).

Given the multiple proposed roles of temporal lobe white matter pathways in reading, we investigated whether a clearer picture might emerge if we considered associations between anatomy and different subcomponents of reading skills separately. On this view, brain–behavior relationships might vary anatomically based on component reading skills, i.e., phonological decoding versus reading comprehension. Here, we explored relationships between white matter anatomy and speeded recognition of real words (sight word efficiency), speeded reading of nonwords (phonemic decoding efficiency), and reading comprehension. These measures were selected to tease apart various aspects of reading skill, and indeed show partially distinct relationships with patterns of brain activation (Welcome and Joanisse, 2012). As such, we explored whether associations with white matter microstructure were specific to these reading subskills. Our study also included a test of nonverbal IQ, which helped identify the extent to which apparent effects went beyond more general differences in global cognitive ability.

It is also apparent that the choice of method of measuring white matter anatomy can influence the nature of the results. Most studies of white matter involvement in reading have to date employed measures of FA and/or voxel-wise coherence using deterministic algorithms. However, there is also growing interest in a somewhat newer approach of probabilistic tractography, which examines the microstructural properties and connectivity of white matter within specific anatomical regions of interest. A challenge of imaging white matter tracts has been the concern that the size of these tracts is appreciably smaller than what can be imaged by conventional MRI scanners, yielding some indeterminacy in identifying the direction of individual fibers being imaged. Earlier work in tractography used deterministic algorithms that provided a “best guess” at fiber orientation. Probabilistic tractography provides some advantage by additionally deriving information regarding the confidence with which a pathway is identified (Dudink et al., 2008). Such algorithms are more resistant to noise as errant paths tend to not be repeated as often as legitimate paths (Behrens et al., 2007). A probabilistic approach also allows tracking of fibers nearer to areas of low anisotropy such as gray matter, rather than being restricted to regions in which fiber direction is more certain (Behrens et al., 2003b; Conturo et al., 1999). These differences may result in improved ability to track temporal lobe pathways. For instance, in one recent study, deterministic tractography identified the right arcuate fasciculus in 34/55 subjects, while probabilistic tractography identified this structure in every subject (Yeatman et al., 2011).

Probabilistic tractography requires the use of anatomical seed regions to serve as either originator and/or waypoint regions. One challenge to this approach is the optimal method for choosing these seed regions. To address this challenge, we used a variety of approaches, including manually-defined cortical ROIs, atlas-based white matter ROIs, and functionally-defined ROIs as seed regions for probabilistic tractography. Using multiple approaches allowed us to investigate the

extent to which these techniques provided converging evidence of relationships between reading ability and white matter anatomy.

Variability in cortical anatomy also makes it difficult to accurately compare white matter tractography among individuals especially when data are transformed to a standard stereotaxic space prior to analysis. Reading group differences have been noted in tract location (Schwartzman et al., 2005) and fiber orientation (Deutsch et al., 2005) within white matter. While transforming data into standardized space is intended to correct for this variability, this approach introduces the possibility that different degrees of anatomical distortions are introduced across individuals, yielding spurious individual differences in measures of white matter diffusion. To address this concern, we present analyses in which each individual's data are analyzed untransformed to a standard stereotaxic space. This approach allowed us to identify differences in white matter connectivity in a way that was not influenced by distortions introduced by spatial registration.

Overview of the present studies

We explored relationships between white matter anatomy and reading ability in typical adults using multiple experimental and analytic approaches. All studies examined the relationship between white matter connectivity and measures of reading subskills, using correlations and multiple regression. However, each experiment involved a different methodology for localizing and quantifying white matter differences. In Experiment 1, we manually delineated cortical ROIs in each hemisphere of each individual brain and performed probabilistic tractography to examine the extent and integrity of white matter pathways projecting from these cortical regions. Experiment 2 investigated cognitive correlates of individual differences in cross-hemispheric connections by using portions of the corpus callosum as seed regions. In Experiment 3, we used standardized, atlas-based white matter pathways transformed into each individual's native space as ROIs to confirm that relationships were not determined by variation in the size and shape of the manually delineated ROIs. Finally, Experiment 4 explored connectivity underlying functional reading networks that were identified using fMRI.

In all cases, we examined individual differences across four cognitive measures related to reading. Timed reading of real words, which might reflect efficiency of word recognition, was assessed with the Sight Word Efficiency subtest of the Test of Word Read Efficiency (TOWRE). Phonological decoding skill was measured by timed reading of pronounceable pseudowords on the TOWRE. Reading comprehension was assessed in order to investigate relationships between brain anatomy and the ability to read for meaning. Finally, nonverbal IQ was measured in order to examine whether white matter relationships were specific to reading, or extended to more general cognitive skill.

General methods

All experiments involved the same participants, cognitive measures and MRI/DTI datasets. However each used a different analytic approach to characterize the relationship between white matter and indices of reading-related subskills.

Participants

Seventeen neurologically normal adults (12 female) took part in the study. All were right-handed native speakers of English recruited from newspaper advertisements in the London, Ontario community and postings at the University of Western Ontario. Demographic data are presented in Table 1. Scores on the reading measures ranged from below average to well above average, representing the variability present in a community sample. As described previously (Welcome and Joanisse, 2012), all participants completed standardized measures of intelligence and reading ability prior to scanning. Nonverbal intelligence

Table 1
Demographic characteristics of the sample (N = 17).

	Mean (SD)	Range
Age	31.8 (14.7)	19–59
TONI (scaled score)	96.3 (18.1)	72–140
TOWRE SWE (scaled score)	99.4 (10.3)	83–113
TOWRE PDE (scaled score)	94.0 (8.0)	84–109
Nelson–Denny Comprehension	216.2 (23.8)	163–251

was assessed with the Test of Nonverbal Intelligence-3 (TONI-3) (Brown et al., 1997). The Sight Word Efficiency (SWE) and Phonemic Decoding Efficiency (PDE) subtests of the Test of Word Read Efficiency (TOWRE) were used to assess timed reading of familiar words and pronounceable pseudowords, respectively (Torgesen et al., 1999). Form G of the Nelson–Denny Reading Comprehension subtest (Brown et al., 1993), in which comprehension questions follow short passages, was used as a timed measure of reading comprehension.

Image acquisition and processing

Imaging was performed on a 3 Tesla Siemens Tim Trio scanner equipped with a 32-channel head coil for transmit/receive. A 3D T1-weighted anatomical scan was acquired (MPRAGE; GRAPPA acceleration factor = 2; TR = 2.3 s, TE = 2.98 ms, flip angle = 9°, matrix = 256 × 240 × 192, voxel size = 1 × 1 × 0.87 mm). Diffusion-weighted brain images were acquired at the end of the scanning session. The acquisition was done in the axial plane using a single-shot echo planar imaging sequence (55 slices with 2 mm slice thickness with no interslice gap, in-plane voxel size = 2.08 × 2.08 mm, matrix = 96 × 96, field of view = 200 mm², 64 diffusion directions with b = 700 s/mm², TR = 6.0 s, TE = 75 ms). These parameters were standard for the imaging facility and chosen to balance concerns regarding sensitivity and signal-to-noise ratio. Across participants, the average maximum displacement was 1.89 mm.

Images were processed using the FSL (FMRIB Software Library, FMRIB, Oxford, UK) (Smith et al., 2004) software package. A brain mask for each participant was created using manual modifications of the mask created by BET (Brain Extraction Tool; Smith, 2002). We performed affine coregistration (Jenkinson and Smith, 2001) to calculate the transformation matrix necessary to align the T1-weighted image and the first diffusion-weighted volume (b0). FDT (FMRIB's Diffusion Toolbox; Behrens et al., 2003a) was used to fit the tensor model and to compute FA maps. Probabilistic tissue-type segmentation was performed on T1-weighted images using FAST (FMRIB's Automated Segmentation Tool; Zhang et al., 2001).

Experiment 1: manually-defined ROIs

We investigated relationships between individual differences in the size and integrity of white matter pathways and typical variation in reading skills among adults. We examined correlations between cognitive measures (speeded word reading, speeded nonword reading, reading comprehension, and nonverbal IQ) and the volume and FA of tracts extending from manually-defined cortical ROIs. Multiple regression analysis was used to examine whether relationships between tract volume and cognitive measures were explained by individual differences in the volume of the manually-drawn ROI.

Regions of interest were selected to include inferior frontal and temporo-parietal regions that are implicated in various reading processes. Inferior frontal cortex, particularly in the left hemisphere, has been found to be differentially active during phonological analysis (Katzir et al., 2005; Pugh et al., 1996; Welcome and Joanisse, 2012). A more anterior portion of the IFG appears to be involved in semantic processing (Gough et al., 2005; McDermott et al., 2003). Temporo-parietal cortex is activated by tasks that place heavy demands on grapheme-to-

phoneme conversion (Blomert and Froyen, 2010; Jobard et al., 2003). The left middle temporal cortex activates more strongly during semantic tasks than phonological or orthographic tasks (Price et al., 1997; Pugh et al., 1996).

Further, these regions were selected because they show altered gray matter anatomy in individuals with impaired reading. For instance, individuals with dyslexia have been reported to show decreased volume in the left temporal lobe as a whole (Brown et al., 2001; Eliez et al., 2000) or in part (Brambati et al., 2004; Eckert et al., 2005; Vinckenbosch et al., 2005). These relationships extend to university students, in whom reduced asymmetry of temporo-parietal gray matter thickness and reduced size of inferior frontal regions were associated with different reading profiles (Welcome et al., 2011).

Analyses

Cortical ROIs were drawn manually in the left and right hemispheres of each individual's anatomical image by three raters blind to behavioral scores. ROIs included superior temporal, middle temporal, Heschl's, angular, and supramarginal gyri. Intra-rater reliabilities (computed as correlations of volumes from a subset of 6 participants) ranged from .89 to .98 across regions, and inter-rater reliabilities ranged from .83 to .96. Briefly, ROIs were defined as follows.

Superior temporal gyrus (STG)

The STG ROI was drawn in the coronal view. The most posterior slice in which the Sylvian fissure was identified served as the posterior boundary, and the ROI continued to the temporal pole at its anterior extent. The Sylvian fissure was the superior boundary, and the superior temporal sulcus (STS) was the inferior boundary. In cases of a discontinuous STS, segments were connected with a straight line.

Middle temporal gyrus (MTG)

The MTG was defined in the coronal view in the same slices as the STG using the STS as the superior boundary and the inferior temporal sulcus as the inferior boundary. When the inferior temporal sulcus branched, the most superior branch served as the inferior boundary.

Heschl's gyrus

Heschl's gyrus was defined in the coronal view. Tracing began with the appearance of Heschl's sulcus, which served as the inferior boundary. The Sylvian fissure was the superior boundary. Tracing continued until the gyrus disappeared into the insular cortex. In the case of duplications, all duplications were included in the ROI.

Angular gyrus

The angular gyrus was defined in the axial view. The primary intermediate sulcus served as the anterior boundary, the posterior branch of

the STS as the posterior boundary, and the primary branch of the STS as the inferior boundary.

Supramarginal gyrus

The supramarginal gyrus was defined in the axial view, using the Sylvian fissure as the inferior boundary, the interparietal sulcus as the superior boundary, the primary intermediate sulcus as the posterior boundary, and the postcentral sulcus as the anterior boundary.

The seed masks included both gray and white matter. Preliminary analyses suggested that tractography results were similar when only gray matter was included (Fig. S1). All ROIs were drawn in native space and transformed into each participant's diffusion-weighted spaced using FLIRT (MRIB's Linear Image Registration Tool; Jenkinson and Smith, 2001). Thus, no inter-subject registration was required and no spatial smoothing was employed.

For each participant, we used probabilistic tractography to track fibers from each voxel in each seed mask (Behrens et al., 2003a,b). For each voxel, 5000 streamlines were sent out, with a steplength of 0.5 mm and a curvature threshold of 80°, and the probability of those streamlines passing through each other voxel in the brain was recorded. To examine tract volume, different thresholds were employed (voxels with a probability greater than the 50th, 75th, 90th, and 95th percentiles) and the volume of tissue exceeding each probability was calculated. Because different thresholds yielded very comparable results, we chose to report the results using the 90th percentile here; additional thresholds are included as Supplementary data (Fig. S1). We correlated both the volume and mean FA of tracts extending from each seed mask with each of our cognitive measures.

To correct for multiple comparisons, a 10,000-iteration permutation test was performed to calculate the chance level of significance given the distribution of FA and tract volumes we have observed here. In each iteration, each subject's tract volume or FA was randomly reassigned to a different subject, and the correlation between cognitive measures and the volume or FA was calculated. If less than 5% (i.e. $p < .05$) of the results from the random tests reached or exceeded the correlation coefficient for the real test, the original relationship was considered significant.

Results

We examined correlations between the volume of tracts extending from each cortical ROI and IQ and reading ability (Table 2). The only reliable association observed was between the volume of the tract extending from left Heschl's gyrus and scores on the phonemic decoding efficiency subtest [$r(15) = .68, p < .005$]. It is worthy of note that the left STG and right Heschl's and STG ROIs yielded similar effects in the same direction, though these effects failed to reach significance. As demonstrated in Fig. 1, larger tract volumes were associated with better nonword reading. Thus, individuals in whom tractography identified

Table 2

Associations (Pearson's r -values) between tract volume extending from manually defined ROIs and psychometric measures.

		Sight word efficiency	Phonemic decoding efficiency	Reading comprehension	Nonverbal IQ
Left hemisphere	Angular gyrus	−0.22	0.02	−0.06	0.22
	Supramarginal gyrus	−0.17	0.31	−0.31	−0.16
	Heschl's gyrus	0.10	0.68***	0.11	0.03
	Superior temporal gyrus	−0.14	0.43	−0.14	−0.09
	Middle temporal gyrus	−0.13	0.21	−0.02	0.01
	Inferior frontal gyrus	−0.01	0.32	0.18	0.27
Right hemisphere	Angular gyrus	−0.29	0.27	−0.07	−0.02
	Supramarginal gyrus	−0.10	0.31	−0.12	−0.02
	Heschl's gyrus	−0.06	0.40	0.19	0.09
	Superior temporal gyrus	−0.02	0.41	0.05	−0.00
	Middle temporal gyrus	0.00	0.20	−0.04	0.04
	Inferior frontal gyrus	−0.03	0.23	0.05	0.17

The bold values have a $p < .05$, not corrected for multiple comparisons.

*** Denotes $p < .005$, corrected for multiple comparisons.

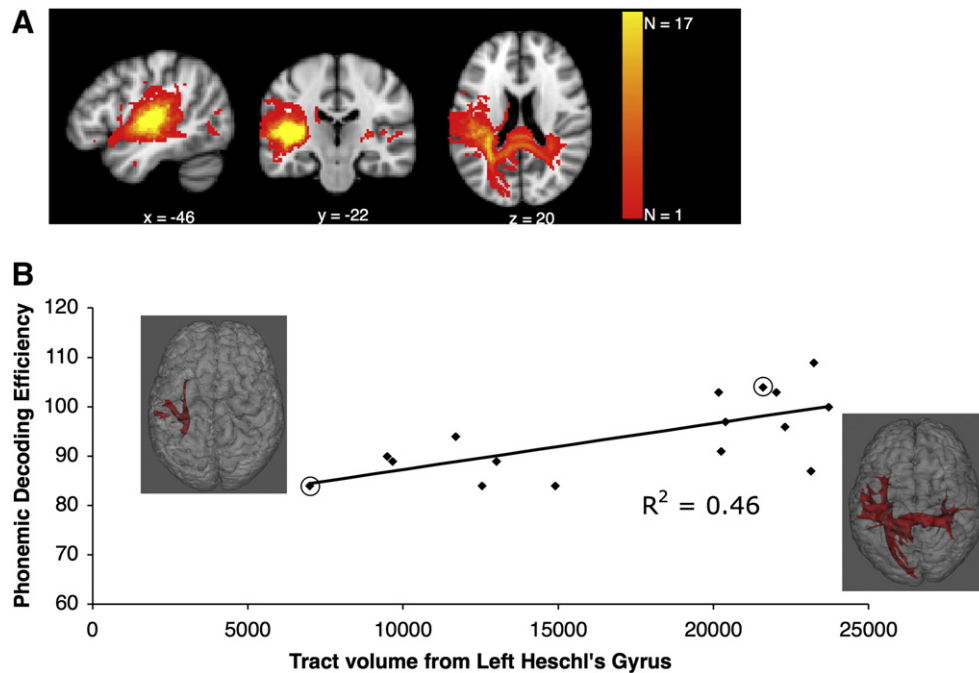


Fig. 1. A) Inter-participant variability in pathways extending from left Heschl's gyrus; color scale denotes the number of participants for which the voxel probability exceeded the 90th percentile displayed over the MNI average brain. B) Volume of tract extending from left Heschl's gyrus predicts scaled score on a measure of nonword decoding (TOWRE Phonemic Decoding Efficiency subtest; $R^2 = 0.46$). 3-Dimensional images of pathways in sample participants with low tract volume and high tract volume illustrate individual differences in pathway size.

more widespread pathways extending from Heschl's gyrus in the left hemisphere tended to perform more strongly on this phonological decoding measure. Multiple regression analysis was used to test whether the relationship between tract volume and PDE was explained by individual differences in the volume of the manually drawn ROI. Even when the volume of the seed region in each individual was partialled out from the volume of the pathways identified by tractography algorithm, volume of the tract extending from left Heschl's gyrus remained a significant and unique predictor of PDE score ($\beta = .57, p < .05$), while volume of the ROI did not ($\beta = .23, p > .10$). Thus, volume of the white matter pathway, rather than the volume of Heschl's gyrus itself, predicted nonword reading ability.

The white matter pathways found extending from Heschl's gyrus include parts of many known white matter structures. In order to divide the white matter underlying Heschl's gyrus into known white matter pathways, the mask representing tissue extending from each participant's left Heschl's gyrus was transformed into MNI space, and regions from a standard white matter atlas (the Johns Hopkins University white-matter tractography atlas as provided within the FSL software package; Mori et al., 2005) were selected when at least 12 participants had portions of their mask extending into the region. In this way, six regions were selected from the atlas, as shown in Table 3. Tracts of interest were transformed into each individual's native anatomical space, and the volume of each tract that overlapped with the pathway extending

from Heschl's gyrus in each participant was calculated. These analyses revealed that individuals with higher PDE scores had larger volumes overlapping with the splenium of the corpus callosum [$r(15) = .52, p < .05$], the retrolenticular portion of the internal capsule [$r(15) = .59, p < .05$], and the posterior corona radiata [$r(15) = .59, p < .05$]. Thus, larger pathways include tracts connecting the two hemispheres and thalamo-cortical projections.

We next examined correlations between cognitive measures and mean FA within identified tracts. As shown in Table 4, significant associations were observed between PDE and FA in the tract extending from left Heschl's gyrus [$r(15) = .53, p < .05$], and between nonverbal IQ and FA in the tract extending from the left angular gyrus [$r(15) = .58, p < .05$]. Thus, more coherent tracts underlying left Heschl's gyrus were associated with better phonological decoding performance, while nonverbal IQ was associated with FA within the tract underlying left angular gyrus. This double dissociation is depicted in Fig. 2.

Discussion

Both the volume and FA of white matter underlying primary auditory cortex in the left hemisphere relate to phonological decoding ability, such that larger and more coherent pathways were associated with more efficient reading of nonwords. Interestingly, these associations did not extend to other reading skills, as neither reading comprehension nor the efficiency of sight word reading was associated with white matter microstructure in this region. We also noted that the left STG, right STG, and right Heschl's gyrus showed non-significant associations with phonological decoding in the same direction, suggesting that subtler effects may extend to superior temporal regions in both hemispheres.

Using a standardized atlas to examine subregions of the tracts projecting from Heschl's gyrus, we found that the greater volume of interhemispheric connections and thalamo-cortical projections was associated with better decoding skill. While some of these do not seem to connect brain regions involved with reading, it is noteworthy that several prior studies using different approaches have also observed

Table 3

Associations (Pearson's r -values) between tract volume that overlap between manually defined left Heschl's gyrus and atlas-based regions and psychometric measures.

	Phonemic decoding efficiency
Splenium of the corpus callosum	0.52*
L retrolenticular part of internal capsule	0.59*
L posterior thalamic radiation	0.09
L external capsule	0.15
L posterior corona radiata	0.59*
L superior longitudinal fasciculus	−0.02

The bold values have a $p < .05$, not corrected for multiple comparisons.

* Denotes $p < .05$, corrected for multiple comparisons.

Table 4Associations (Pearson's *r*-values) between fractional anisotropy in tracts extending from manually defined ROIs and psychometric measures.

		Sight word efficiency	Phonemic decoding efficiency	Reading comprehension	Nonverbal IQ
Left hemisphere	Angular gyrus	0.14	−0.19	0.43	0.58*
	Supramarginal gyrus	0.19	0.27	0.03	0.07
	Heschl's gyrus	0.02	0.53*	0.16	−0.08
	Superior temporal gyrus	0.11	0.47	−0.09	−0.10
	Middle temporal gyrus	−0.06	0.06	−0.10	−0.05
	Inferior frontal gyrus	0.10	0.25	0.02	−0.10
Right hemisphere	Angular gyrus	−0.09	0.23	0.13	0.10
	Supramarginal gyrus	−0.02	0.40	−0.05	−0.01
	Heschl's gyrus	0.04	0.30	0.43	0.17
	Superior temporal gyrus	0.19	0.31	0.04	−0.11
	Middle temporal gyrus	0.00	0.01	−0.01	−0.07
	Inferior frontal gyrus	−0.01	0.20	−0.01	−0.07

The bold values have a $p < .05$, not corrected for multiple comparisons.* Denotes $p < .05$, corrected for multiple comparisons.

associations between anatomy of the corona radiata and reading ability (Beaulieu et al., 2005; Niogi and McCandliss, 2006; Odegard et al., 2009). In this experiment, the fibers were first identified as part of a pathway extending from primary auditory cortex, and only later divided into white matter structures. Thus, while the location of the white matter effects may be surprising, the fibers in question in this experiment have their origin in primary auditory cortex.

From Fig. 1, it is apparent that one factor contributing to individual differences in tract volume extending from left Heschl's gyrus is individual differences in callosal anatomy. Those individuals with larger pathways from Heschl's gyrus showed cross-callosal projection, while such projections were not identified in individuals with the smallest tracts. With this in mind, we next examined correlations between the volume of tracts extending through each callosal ROI and the cognitive measures of interest.

Experiment 2: callosal ROIs

Prior work has demonstrated mixed associations between anatomy of the corpus callosum and reading ability. Better phonological awareness skill was associated with lower FA and higher mean diffusivity in a portion of the callosum projecting to the temporal lobes (Dougherty et al., 2007). Worse reading of both words and pseudowords was associated with greater FA within the posterior corpus callosum (Odegard et al., 2009). Dyslexic readers show higher FA in the splenium, but phonological awareness was positively associated with FA in the splenium among typical readers (Frye et al., 2008).

Our own results add to this prior literature and suggested an association between callosal anatomy and nonword reading skill. Specifically, the results of Experiment 1 suggest that the tracts that were larger in individuals with higher PDE scores had regions of overlap with the splenium of the corpus callosum. To further investigate this association, we manually delineated portions of the corpus callosum (Fig. 3) in each participant according to standard methods, and then used probabilistic tractography to identify the size and extent of fiber tracts projecting through each of these regions.

Analyses

Because tract volume differences across individuals seemed to be partially explained by variation in the volume of cross-hemispheric connections, we explored relationships between volume and FA within callosal pathways and cognitive measures. The corpus callosum was segmented subjectwise within native (non-stereotaxic) space on anatomical images by three raters blind to behavioral scores following the Witelson (1989) parcellation scheme. Probabilistic tractography was performed using the same parameters as laid out in Experiment 1 and the same correction for multiple comparisons was used.

Results

As indicated in Table 5, larger tracts passing through the rostrum [$r(15) = .63, p < .01$] and isthmus [$r(15) = .56, p < .05$] of the corpus callosum were significantly correlated with phonemic decoding ability (correlations in the rostral body and posterior midbody also approached, but did not reach, statistical significance). Thus, as shown in Fig. 4, more cross-hemispheric fibers in the rostrum and isthmus were identified in individuals with higher phonemic decoding efficiency scores. A potential concern is that these results stem strictly from the actual size of these portions of an individual's corpus callosum, rather than the relative coherence of white matter tracts projecting through it. To test this, the volume of the seed region in each individual was partialled out from the volume of the pathways identified by tractography algorithm; we observed that the volume of the tract through the isthmus continued to make a significant, unique prediction of PDE score ($\beta = .57, p < .05$), while cross-sectional area of the seed ROI did not ($\beta = -.03, p > .10$). However, the picture was less clear for the rostrum, where the volume of the tract did not make a unique prediction of PDE score ($\beta = .40, p > .10$), nor did cross-sectional area ($\beta = .38, p > .10$). The data suggest that tract extent is at least partially independent of the size of the tract itself, and that it captures at least a portion of variance in PDE score that is not apparent from the basic anatomical size of a white matter tract.

We also explored relationships between FA and cognitive measures. Higher FA within tracts extending through the anterior midbody was predictive of higher reading comprehension scores [$r(15) = .60, p < .05$] (Table 6). Unlike the tract volume findings, no reliable relationships were observed with phonemic decoding ability.

Discussion

Based on the observation that individual differences in tract volume from left Heschl's gyrus might relate to volume of cross-hemispheric connections, we explored relationships between callosal microstructure and reading ability. Tract volume through the isthmus and rostrum was associated with nonword reading skill such that individuals with more extensive cross-callosal connections showed more efficient phonemic decoding. In the isthmus, but not the rostrum, this relationship was independent of the seed region size. While the majority of temporal fibers project through the splenium, some temporal fibers project through the isthmus (Park et al., 2008; Westerhausen et al., 2009). Thus, it is possible that the isthmus finding relates directly to the findings of Experiment 1, and suggests that greater connectivity of left primary auditory cortex, particularly greater cross-callosal connectivity, is associated with better nonword reading. Subsequent studies examining associations between tract destination and reading ability would be useful in order to more fully investigate this relationship.

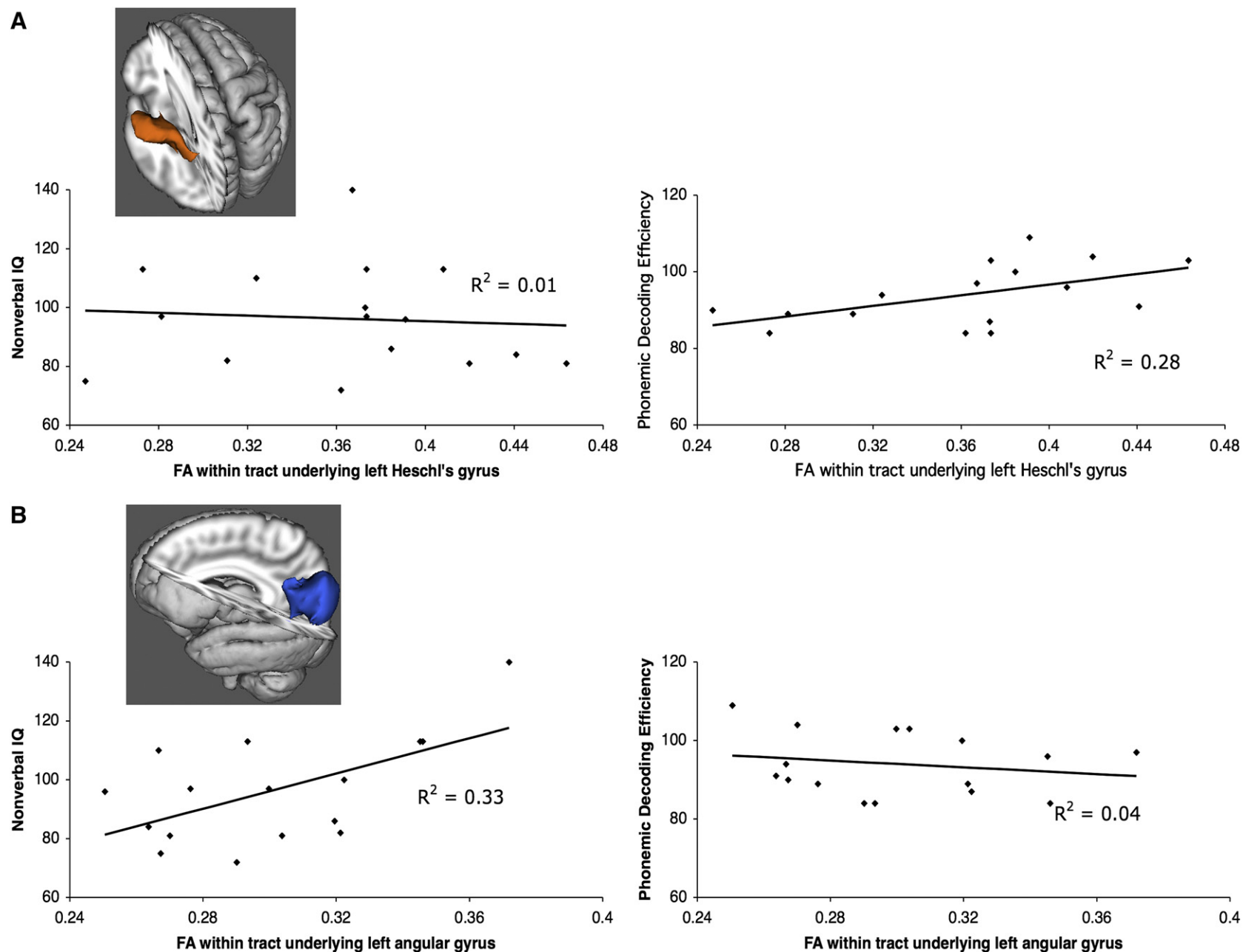


Fig. 2. Double dissociation in relationships between regional fractional anisotropy (FA) and psychometric measures. A) FA in pathways extending from left Heschl's gyrus predicts scaled score on nonword reading (TOWRE Phonemic Decoding Efficiency subtest; $p < .05$) but not nonverbal IQ (TONI; $p = \text{NS}$). ROI representing pathway overlap in eight or more participants (at least half of the sample) is shown in orange. B) FA in pathways extending from left angular gyrus predicts nonverbal IQ (NS) but not scaled score on nonword reading ($p < .05$). ROI representing pathway overlap in eight or more participants (at least half of the sample) is shown in blue.

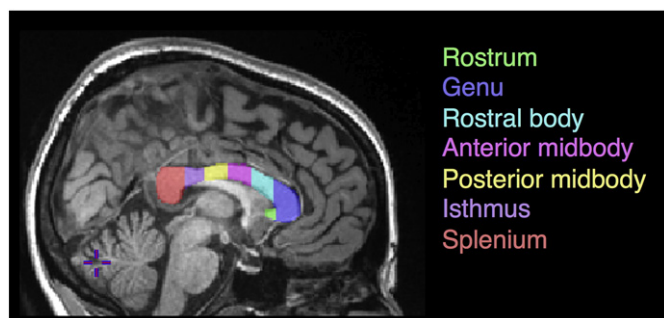


Fig. 3. Callosal ROIs drawn for a representative individual following the Witelson (1989) parcellation scheme.

Greater FA within the tract extending through the anterior midbody was associated with better reading comprehension. Fibers passing through the anterior midbody typically have their origins in the pars opercularis, superior frontal, and caudal middle frontal regions (Pannek et al., 2010). Bilateral activation in these frontal regions has been associated with reading for meaning (Moss et al., 2011; Welcome and Joanisse, 2012). Therefore, efficient reading comprehension may rely on greater integrity of fibers passing through the anterior midbody of the corpus callosum.

Experiment 3: atlas-based ROIs

In Experiment 1, cortical ROIs drawn for each individual participant revealed that the size and integrity of tracts underlying left Heschl's gyrus predicted phonological decoding ability. However, these ROIs and resulting tracts have different sizes and shapes in each individual, and therefore results from Experiment 1 could reflect individual differences in the size and location of tracts identified by probabilistic tractography. Therefore, to examine FA within tracts of the same size and shape in each individual, tracts of interest were selected from a standardized white matter atlas (Mori et al., 2005) and transformed into each individual's native anatomical space. This approach had the further advantage of allowing the investigation of FA within specific white matter pathways associated with language, including the inferior longitudinal fasciculus, superior longitudinal fasciculus, uncinate fasciculus, and inferior fronto-occipital fasciculus (Catani and Mesulam, 2008; Duffau et al., 2005; Mandonnet et al., 2007). We were thus able to tease apart FA within different pathways passing through the temporal lobe in order to examine correlations with component reading skills. By identifying tracts of interest, rather than taking the traditional approach of correlating FA at each voxel with cognitive measures, we were able to minimize the likelihood of type I errors due to large numbers of statistical comparisons.

Analyses

While manually specified anatomical ROIs have advantages, the ROIs and resulting tracts will have different sizes and shapes based on

differences in each individual's brain morphology. It is unclear whether such effects reflect differences in neural processing, or are simply an innocuous artifact of individual variations in regional brain morphology. To examine whether differences in FA are also evident when ROIs are matched for size and shape across individuals, tracts of interest were selected from a standardized probabilistic tractography atlas (the Johns Hopkins University white-matter tractography atlas as provided within the FSL software package; Mori et al., 2005). Binary masks thresholded at a probability of 10% were transformed from standard space into each participant's DTI space by inverting the affine transformation necessary to align the native space brain to MNI space. We chose to examine FA within the superior longitudinal fasciculus, inferior longitudinal fasciculus, uncinate fasciculus, and inferior fronto-occipital fasciculus in each hemisphere of each participant. As in the other experiments, we used permutation testing to correct for multiple comparisons.

Results

We examined correlations between FA in the atlas-based tracts of interest and our four cognitive measures (Table 7). The only reliable associations observed involved phonemic decoding ability. Higher FA within the inferior fronto-occipital fasciculus in both the left [$r(15) = .51, p < .05$] and right hemispheres [$r(15) = .64, p < .01$], and within the left uncinate fasciculus [$r(15) = .50, p < .05$] was associated with better performance on the PDE measure (Fig. 5). These results suggest that the effect is driven by variation in FA within fronto-temporal connections and fronto-occipital connections that pass through the temporal lobe, as relationships were strongest in the uncinate fasciculus and fronto-occipital fasciculus. We also noted that a negative relationship with nonverbal IQ and the left and right inferior longitudinal fasciculus fell just short of statistical significance.

In order to more fully explore potential relationships between reading ability and perisylvian white matter anatomy, we performed similar analyses using the Natbrainlab Atlas of the perisylvian network. This atlas provides ROIs in standard space that represent the long segment, and posterior and anterior segments of the arcuate. No relationship with FA within any of these regions correlated significantly with PDE. In fact, only one relationship, between FA in the posterior segment in the left hemisphere and reading comprehension, was statistically significant [$r(15) = .51, p < .05$].

Discussion

Our first two experiments included analyses of white matter tracts using ROIs defined manually for each individual. That approach helps guard against effects of spatial transformations used to align each individual to a common anatomical template. However, it also introduces the possibility that differences in the size and shape of seed ROIs determine differences in tract properties. Experiment 3 used a different approach: we employed a standardized anatomical atlas of white matter tracts to isolate specific ROIs of interest. Clear relationships with phonemic decoding efficiency emerged. Higher FA within the inferior fronto-occipital fasciculus bilaterally and the left uncinate fasciculus

Table 5
Associations (Pearson's r -values) between tract volume extending from callosal ROIs and psychometric measures.

	Sight word efficiency	Phonemic decoding efficiency	Reading comprehension	Nonverbal IQ
Rostrum	0.10	0.63**	0.15	0.08
Genu	0.08	0.28	0.27	0.16
Rostral body	−0.03	0.47	0.23	−0.03
Anterior midbody	−0.16	0.37	−0.15	−0.26
Posterior midbody	−0.17	0.43	−0.24	−0.26
Isthmus	−0.03	0.56*	−0.16	−0.14
Splenium	−0.01	0.30	0.08	0.05

The bold values have a $p < .05$, not corrected for multiple comparisons.

* Denotes $p < .05$, corrected for multiple comparisons.

** Denotes $p < .01$, corrected for multiple comparisons.

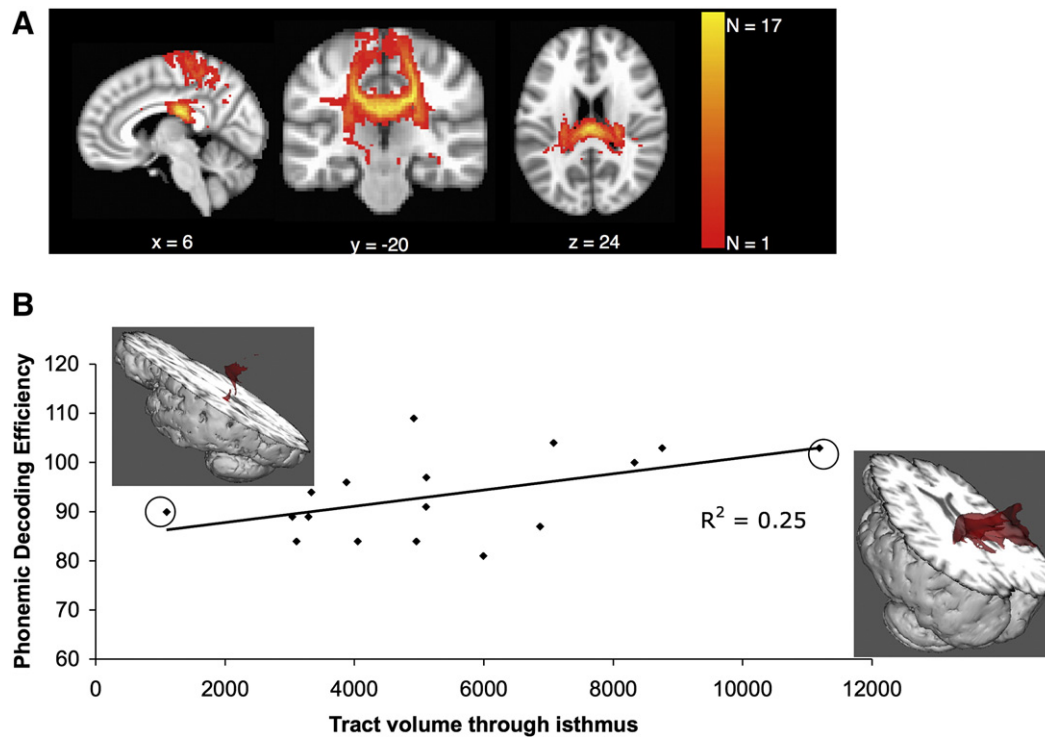


Fig. 4. Associations between corpus callosum white matter tracts and reading. A) Inter-participant variability in tract volume with the isthmus as the seed region; color scale denotes the number of participants for which the voxel probability exceeded the 90th percentile displayed over the MNI average brain. B) Volume of tract extending through the isthmus of the corpus callosum predicts scaled score on the TOWRE Phonemic Decoding Efficiency subtest ($R^2 = 0.25$). 3-Dimensional images of pathways in a participant with low tract volume and a participant with high tract volume illustrate individual differences in pathway size.

was associated with better nonword reading. Prior studies suggest that these regions represent key sites of transmission between frontal and temporal (uncinate fasciculus) and frontal and occipital (frontal-occipital fasciculus) regions.

Prior work has demonstrated associations between FA and the left arcuate fasciculus and phonological skill in children (Yeatman et al., 2011, 2012). The present study used a different measure of phonological ability (speeded nonword reading) from prior work, which has used phonological awareness (Yeatman et al., 2011) or a composite measure that includes sight word identification (Yeatman et al., 2012). It is possible that differences in task demands explain the divergent findings. In adults, phoneme awareness and speech perception tasks, but not rapid word identification, were linked to FA in the arcuate fasciculus (Vandermosten et al., 2012), indicating that perhaps these results relate to spoken language manipulation and perception. The nonword reading measure used here requires print-to-sound conversion in addition to knowledge of speech sounds, and might therefore show different associations with anatomy.

Disruption of both the uncinate fasciculus and inferior fronto-occipital fasciculus has been associated with altered language production (Grossman et al., 2004; Mandonnet et al., 2007). Further, functional imaging studies have highlighted the key role of inferior frontal (Katzir

et al., 2005; Pugh et al., 1996) and temporal (Pugh et al., 1996) regions in phonological processing. Temporo-parietal cortex may be involved in grapheme-to-phoneme conversion (Blomert and Froyen, 2010), while inferior frontal activation might reflect access to speech sound representations (Zatorre et al., 1996). In this light, it is interesting to note that those individuals with greater connectivity between these regions engaged by phonological decoding perform better on the nonword reading measure. Better nonword reading was also associated with greater connectivity between frontal and occipital regions, perhaps reflecting the role of occipital regions in the kind of complex visual analysis that reading requires. Thus, it appears that greater connectivity between frontal cortex and occipital and temporal regions predicts phonological decoding ability.

That said, it can be difficult to interpret the role that white matter tracts play in reading-related processes based solely on prior studies of functional activation. A wide range of brain regions have been identified during reading, and there remain many unresolved questions about the specific role that each of these plays in different component processes of visual word recognition. In a previous study, we identified regions of cortical activation during reading tasks using fMRI, in the same participants being examined here (Wellcome and Joanisse, 2012). In Experiment 4, these different regions of activation have been used as seed

Table 6

Associations (Pearson's r -values) between fractional anisotropy in tracts extending from callosal ROIs and psychometric measures.

	Sight word efficiency	Phonemic decoding efficiency	Reading comprehension	Nonverbal IQ
Rostrum	0.08	0.27	0.31	0.22
Genu	0.04	−0.14	0.18	−0.03
Rostral body	0.21	0.30	0.20	−0.09
Anterior midbody	0.37	0.31	0.60*	0.42
Posterior midbody	0.18	−0.08	0.29	0.19
Isthmus	0.23	−0.16	0.45	0.36
Splenium	−0.02	0.04	−0.19	−0.36

The bold values have a $p < .05$, not corrected for multiple comparisons.

* Denotes $p < .05$, corrected for multiple comparisons.

Table 7Associations (Pearson's r -values) between fractional anisotropy in tracts extending from atlas-based and psychometric measures.

		Sight word efficiency	Phonemic decoding efficiency	Reading comprehension	Nonverbal IQ
Left hemisphere	Inferior longitudinal fasciculus	−0.15	0.12	−0.03	−0.47
	Superior longitudinal fasciculus	0.04	0.28	0.33	0.05
	Inferior fronto-occipital fasciculus	0.06	0.51*	0.19	−0.25
	Uncinate fasciculus	0.33	0.50*	0.24	0.12
Right hemisphere	Inferior longitudinal fasciculus	−0.05	0.28	−0.12	−0.47
	Superior longitudinal fasciculus	0.05	0.19	0.03	−0.12
	Inferior fronto-occipital fasciculus	0.14	0.64***	0.05	−0.16
	Uncinate fasciculus	0.24	0.43	0.02	−0.18

The bold values have a $p < .05$, not corrected for multiple comparisons.* Denotes $p < .05$, corrected for multiple comparisons.*** Denotes $p < .005$, corrected for multiple comparisons.

regions for white matter tractography, allowing us to test how white matter connectivity relates to individual differences in specific subcomponents of reading.

Experiment 4: functional ROIs

Probabilistic tractography based on anatomical ROIs has provided evidence that size and integrity of some white matter tracts are associated with phonological decoding, a specific reading subskill. We next explored connectivity underlying functional reading networks engaged by phonological and semantic processing. We used previously obtained functional imaging data from the same participants to identify subnetworks of brain regions associated with different component processes in reading. The resulting functional ROIs allowed us to identify functional networks engaged by specific reading subskills, and then to examine white matter pathways underlying these cortical networks. Each ROI was unique to each individual. This yielded the additional benefits of taking into account different sizes, locations and extents of activated gray matter across participants, as well as allowing us to analyze each participant's anatomy within their native (untransformed) space.

Analyses

fMRI data from each individual was used to identify reading-relevant functional ROIs, which were in turn used as seed masks for probabilistic tractography. Functional data were drawn from a prior study in which we examined fMRI activation during different word recognition tasks (Welcome and Joanisse, 2012). Briefly, participants performed orthographic (case), phonological (rhyme), and semantic (animal/non-animal) judgments on pairs of letter strings in alternating blocks of trials. The contrasts of interest in the current study identified the entire reading network (case, rhyme, and meaning judgments > rest baseline), regions sensitive to phonological processing (rhyme judgments > case judgments), and regions sensitive to semantic processing (meaning judgments > rhyme/case judgments). For each individual, ROIs were created by isolating voxels with activations below a threshold of $p > .001$ for each contrast. To correct for multiple comparisons, permutation testing was used.

Results

The first analysis used voxels engaged by all reading tasks compared to resting baseline as a seed region for probabilistic tractography. Neither the spatial extent nor FA of white matter pathways underlying the reading network as a whole was related to the cognitive measures of interest (Tables 8 and 9). When the functional contrast focused on regions engaged specifically by phonological analysis (rhyme judgments > case judgments), relationships with both nonverbal IQ [$r(15) = .55, p < .05$] and phonemic decoding efficiency [$r(15) = .50, p < .05$] were observed. Thus, those individuals who had larger white matter pathways underlying their “phonological” network performed better on the measure of phonological decoding (Fig. 6A) and nonverbal IQ. When the volume of

the seed region in each individual was partialled out from the volume of the pathways identified by tractography algorithm, volume of the tract made a significant, unique prediction of PDE score ($\beta = .54, p < .05$), while volume of the functional seed region did not ($\beta = .09, p > .10$). An analysis of this network broken into subcomponents on the basis of atlas regions is presented as supplemental data (Table S1).

Conversely, when the functional contrast identified regions that were more engaged by semantic analysis (animal/non-animal judgments > case and rhyme judgments), associations with tract volume were instead reliable for the reading comprehension measure [$r(15) = .59, p < .05$] (Fig. 6B; Tables 8–9). When the volume of the seed region in each individual was partialled out from the volume of the pathways identified by tractography algorithm, volume of the tract significantly contributed to the prediction of reading comprehension score ($\beta = .65, p < .05$), while volume of the functional seed region did not ($\beta = .32, p > .10$). An analysis of this network broken into subcomponents on the basis of atlas regions is presented as supplemental data (Table S2). Thus, dissociations in functional networks engaged by different reading tasks were mirrored in different relationships between cognitive measures and white matter connectivity measures.

Discussion

Experiment 4 investigated connectivity underlying functional reading networks identified with fMRI. For the contrast that identified the “phonological” network, there were reliable associations with phonemic decoding ability and nonverbal IQ. The task that was used to define the ROI, rhyme judgment, was selected to place demands on phonological processing. We previously demonstrated that greater activation in the left inferior parietal lobule and left middle temporal gyrus during word reading was associated with better nonword reading. Here, we extend those findings to suggest that greater connectivity between regions engaged by phonological processing supports more efficient phonological decoding. The contrast that identified the “semantic” network, meaning judgments versus case and rhyme judgments, was selected to identify networks that process word meanings. Greater connectivity between regions engaged by semantic processing is associated with higher reading comprehension scores. This dissociation suggests that different aspects of reading relate to partially distinct aspects of white matter microstructure. The relationships with tract size were not modulated by the size of the functional seed regions. Thus, it is not the size of the cortical region activated by the task, but the size of the white matter network underlying the functional network, that was related to the cognitive measures of interest.

General discussion

This work is the first to investigate correlations between component reading skills and white matter pathway volume and integrity in individual adult readers. Such relationships were apparent within specific pathways and restricted to specific reading skills. White matter tracts were imaged using diffusion tensor imaging, and subsequently analyzed

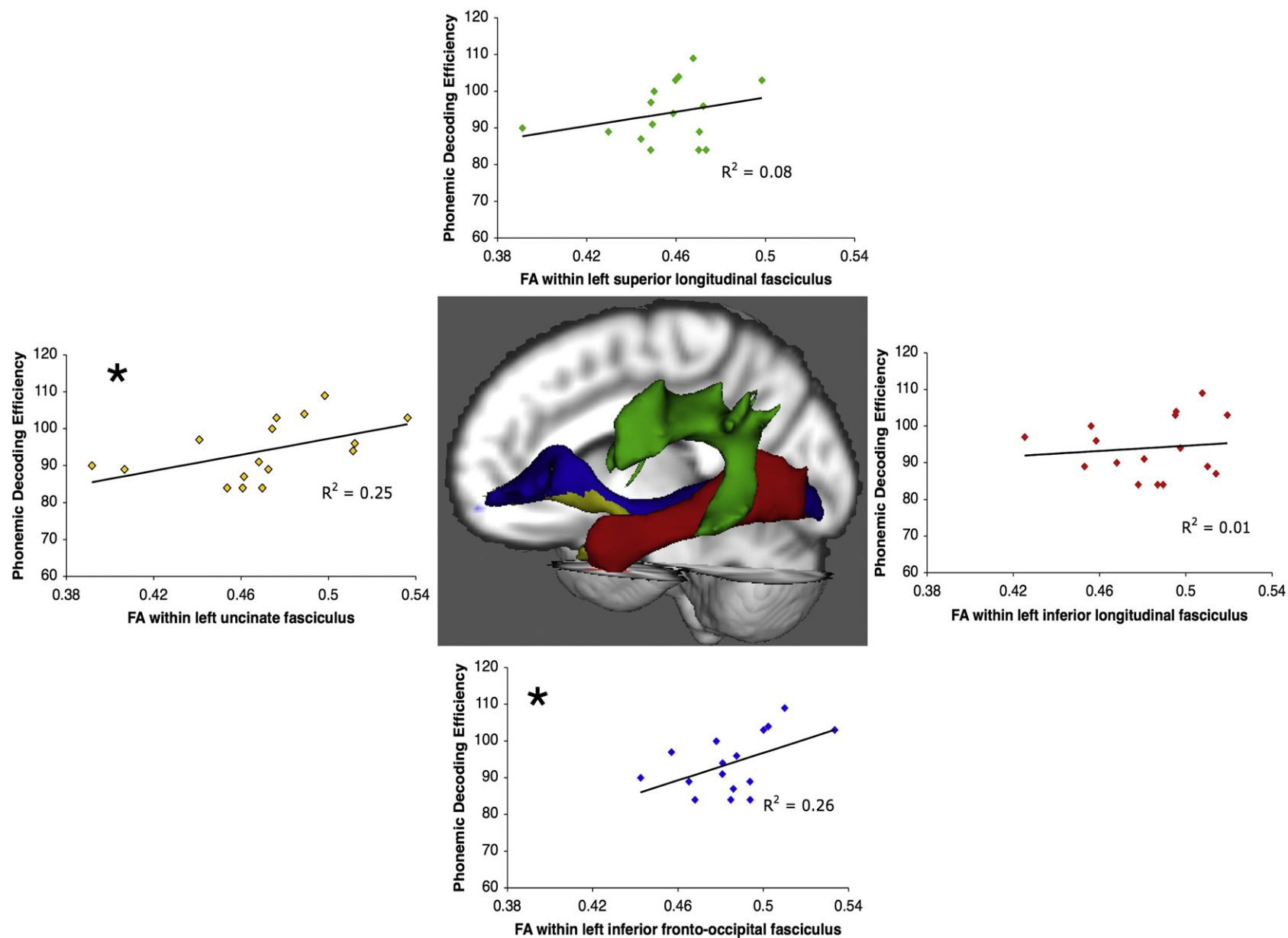


Fig. 5. Associations between fractional anisotropy (FA) within atlas-based ROIs and scaled score on the nonword reading test (TOWRE Phonemic Decoding Efficiency). Significant associations are denoted with an asterisk.

Table 8
Associations (Pearson's *r*-values) between tract volume extending from functional ROIs and psychometric measures. *p*-Values from permutation corrections for multiple comparisons are shown in parentheses.

	Sight word efficiency	Phonemic decoding efficiency	Reading comprehension	Nonverbal IQ
Task > rest	0.07	0.28	0.30	0.37
Rhyme > case	0.17	0.50	0.27	0.55*
Meaning > case/rhyme	0.47	0.30	0.59**	0.47

The bold values have a $p < .05$, not corrected for multiple comparisons.

* Denotes $p < .05$, corrected for multiple comparisons.

** Denotes $p < .01$, corrected for multiple comparisons.

using multiple approaches: manually-defined white matter regions, standard-atlas tracts, and functional ROIs. The results from each of these yielded comparable results, marked by greater fiber integrity and volume of left temporal white matter pathways associated with increased phonemic decoding efficiency in reading. The principal finding is that volume and integrity of white matter underlying Heschl's gyrus in the left hemisphere relate to phonological decoding ability, such that larger and more coherent pathways are associated with better non-word reading. Using standardized, atlas-based ROIs (Experiment 3), we also demonstrated that this relationship is apparent in the inferior fronto-occipital fasciculus and uncinate fasciculus that connect occipital and anterior temporal cortex with inferior frontal cortex. Additionally, more extensive cross-hemispheric connections through the isthmus of the corpus callosum were associated with better phonological decoding (Experiment 2). Connectivity underlying a functional ROI that was more active during phonological processing also related to decoding ability, while connectivity underlying a functional ROI that was more active during semantic processing related to reading comprehension ability (Experiment 4).

Our results relate to prior studies identifying regions of association between white matter microstructure and reading skill (Beaulieu et al., 2005; Deutsch et al., 2005; Klingberg et al., 2000; Niogi and McCandliss, 2006; Odegard et al., 2009). These prior studies have used a variety of methods, including voxel-based correlative analyses of all white matter, ROIs selected using the Reproducible Objective Quantification Scheme, and Tract Based Spatial Statistics. Despite variations in the methods used, these studies, and the experiments presented here, provide converging evidence that reduced integrity of white matter underlying temporal cortex is associated with reduced reading skill. The fiber orientations identified in prior studies are inconsistent, and it is therefore difficult to determine what the cortical projections of these regions might be. Using probabilistic tractography to identify pathways extending from manually defined cortical ROIs in Experiment 1, we noted associations with reading skill in pathways underlying primary auditory cortex. Thus, it is possible that regions of association between FA and reading noted in prior studies contain projections to auditory cortex.

This pattern of results speaks to the white matter network that is implicated in mapping orthography to speech sounds. Individuals with more extensive white matter networks underlying left Heschl's gyrus and passing through the isthmus of the corpus callosum performed better on a standardized measure of phonological decoding, suggesting that greater interconnectivity of brain regions had beneficial effects on an individual's efficiency of converting print to sound. Surprisingly, the relationship between white matter properties and phonological decoding ability was stronger in ventral pathways (uncinate and inferior frontal-occipital fasciculi) than in the superior longitudinal fasciculus, which has shown prior associations with phonological tasks (Yeatman

et al., 2011) and alterations with dyslexia (Carter et al., 2009). The association we observed suggests that ventral pathways, connecting inferior frontal cortex with occipital and anterior temporal regions, support print-to-sound conversion processes or other aspects of phonological processing. This interpretation broadens the role of these ventral tracts beyond their previously proposed involvement in semantic processing (Duffau et al., 2005; Saur et al., 2008). These associations may indicate that greater connectivity between occipital regions, engaged by complex visual analysis, and inferior frontal regions, engaged in phonological analysis, allows for more efficient decoding of pronounceable stimuli.

Nonverbal IQ provided additional insight into the specificity of associations between reading skills and white matter anatomy. We noted that higher nonverbal IQ was associated with higher FA in the tract extending from the left angular gyrus (Experiment 1). Prior studies have also found associations between nonverbal intelligence and FA in white matter underlying parietal cortex, among other regions. Specifically, in children, IQ scores were positively correlated with FA in the left arcuate fasciculus, bilateral frontal regions, and right parietal and temporo-parietal areas (Schmithorst et al., 2005). In adults, greater spatial intelligence was associated with greater FA in the right superior corona radiata (Tang et al., 2010). Additionally, FA measures were correlated with non-verbal intelligence in the cingulum, optic radiation, superior fronto-occipital fasciculus, internal capsule, callosal isthmus, and the corona radiata in adults (Chiang et al., 2009). We note that our findings are broadly consistent with these previous DTI findings, though we cannot pinpoint the specific pathways responsible for the association in our study.

Relationships between phonological decoding and white matter tracts extending from auditory cortex did not appear to also extend to a similar measure of speeded real-word recognition (Sight Word Efficiency). This suggests that specific differences in processing demands between the two measures, specifically explicit print-to-sound conversion and related phonological processing abilities, were the strongest determinant of individual differences in the white matter properties that we investigated. As such, we suggest that greater volume and FA in tracts extending from primary auditory cortex relate specifically to increased efficiency of translating between visual and phonological codes. In this light, it is interesting to note prior findings that activation of early auditory areas was shown to be modulated by simultaneous presentation of visual letters, and show sensitivity to the congruency of letter-speech sound pairs (van Atteveldt et al., 2004). Accordingly, we suggest that auditory cortex, and the white matter underlying it, may have a role in processing associations between letters and their corresponding sounds.

Many of the associations we noted were specific to phonological decoding. In fact, we noted no reliable associations between white matter anatomy and sight word efficiency. It is possible that sight word

Table 9
Associations (Pearson's *r*-values) between fractional anisotropy in tracts extending from functional ROIs and psychometric measures.

	Sight word efficiency	Phonemic decoding efficiency	Reading comprehension	Nonverbal IQ
Task > rest	−0.18	0.01	0.21	−0.05
Rhyme > case	0.13	0.32	0.01	−0.07
Meaning > case/rhyme	0.03	−0.35	−0.04	−0.24

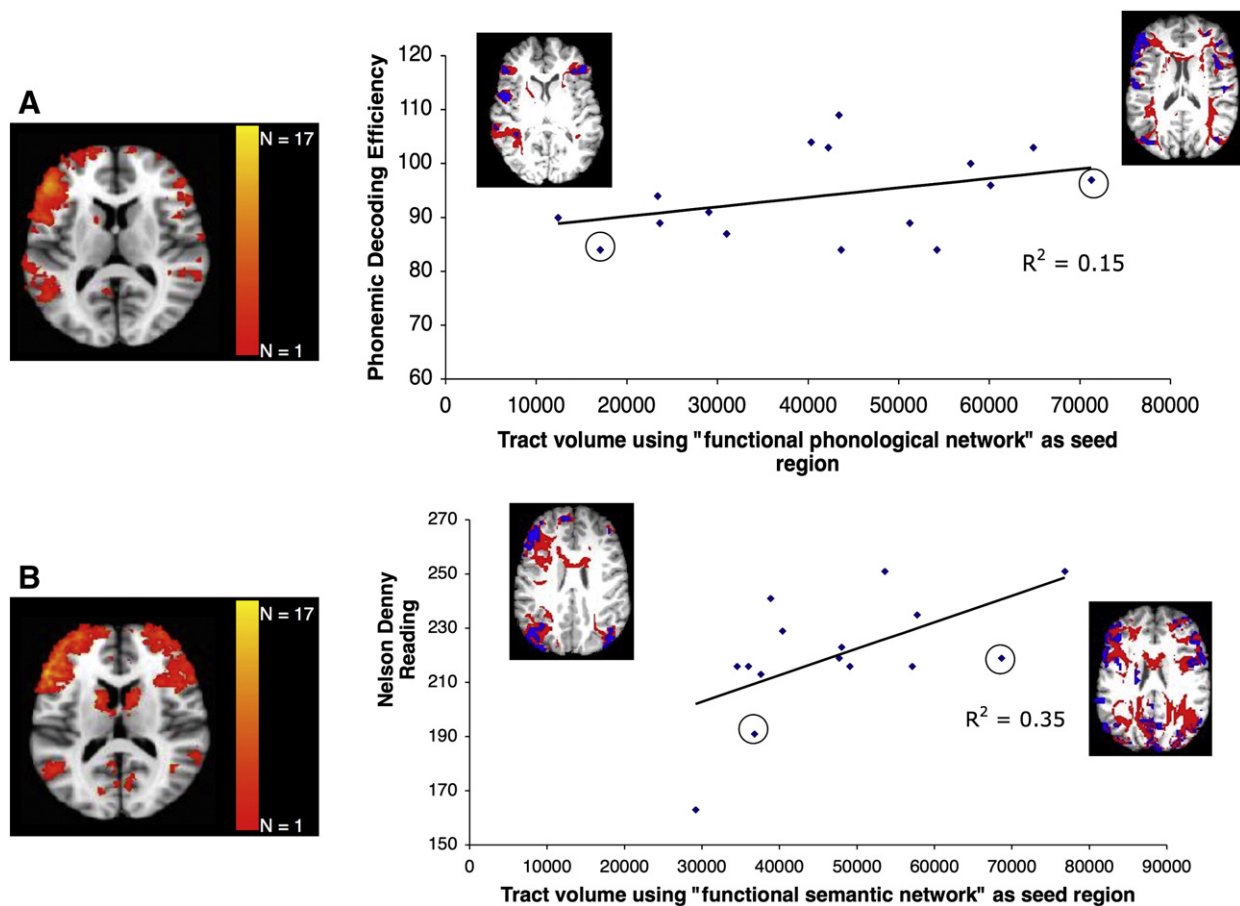


Fig. 6. A) Larger white matter networks underlying regions associated with phonological processing predict higher scaled scores on the TOWRE Phonemic Decoding Efficiency subtest. Functional ROIs, shown in blue, indicate regions showing greater activation for rhyme > case decisions in individual participants. Probabilistic tractography performed using those regions as seeds, shown in red, in participants with lower and higher tract volumes, circled. B) Larger white matter networks underlying regions associated with semantic processing predict higher scaled scores on the Nelson–Denny Reading Comprehension subtest. Functional ROIs, shown in blue, indicate regions showing greater activation for meaning > case/rhyme decisions in individual participants. Probabilistic tractography using those regions as seeds (shown in red) demonstrates variability in tract volume.

recognition relies less critically on interconnectivity with temporal regions than other reading skills, particularly nonword reading. Thus, it is possible that for skills like sight word reading, variance between individuals is driven primarily by processing differences within a single cortical region, whereas for phonological decoding, individual differences result from connectivity between a network of regions. This suggests that efficient reading of nonwords requires integrating multiple distinct cortical regions, reflected in more widespread tracts with greater integrity in those individuals with more efficient phonological processing skills. In contrast, holistic word reading strategies, such as familiar word reading in skilled adult readers, might not rely as strongly on connectivity, particularly in the temporal regions under investigation here. In keeping with this explanation, our earlier functional imaging data have shown that individual differences in phonological decoding correlated positively with reading-related activation in both temporal and other cortical regions, while sight word efficiency correlated with activation primarily within occipital regions (Welcome and Joanisse, 2012). Future research will be necessary to elucidate associations between word reading efficiency and white matter anatomy in the occipital lobe.

The specificity of relationships between white matter anatomy and different reading abilities was underscored by analyses that used functional networks as seed regions (Experiment 4). We used functional ROIs that identified white matter networks underlying cortical networks engaged by phonological and semantic processes. These analyses revealed a double-dissociation — the volume of the network underlying phonological processing predicted phonemic decoding efficiency, and

the volume of the network underlying semantic processing predicted reading comprehension. This finding further supports the hypothesis that different white matter pathways are involved in different component subskills of reading.

One open question worth mentioning concerns whether individual differences in white matter that we and others have identified are a cause or a consequence of individual reading ability. On one view, variation in overall reading ability and its component processes stem from subtle differences in white matter integrity present from an early age. On this view white matter coherence is relatively immutable and places a strong limit on the effect of experience. So for instance, work by Hoeft et al. (2011) have found differences in the extent to which children with dyslexia respond to intervention as a function of white matter organization within right superior longitudinal fasciculus. Such a finding suggests that such characteristics of white matter tracts represent a limiting factor on reading success. A different view holds that the integrity of white matter tracts can change as a function of experience (Yeatman et al., 2012). From this perspective, individual differences in white matter reflect, at least to some extent, differences in environmental factors such as reading experience and learning and teaching styles. Indeed there is some evidence of changes in white matter as a result of reading intervention (Keller and Just, 2009). Given findings in both directions, it is likely that the white matter variability observed in the present study stems both from endogenous sources and environmental factors. That said, longitudinal studies of MRI and behavioral development spanning early childhood through adolescence may provide valuable insights into this question.

Conclusions

We examined associations between different reading abilities and the size and FA of temporal lobe white matter pathways in typical adult readers. Anatomical ROIs defined manually, standardized anatomical ROIs, and functional ROI yielded a similar pattern of results, namely that nonword reading ability was predicted by larger temporal pathways with greater fiber integrity. Such findings suggest that greater connectivity underlying primary auditory cortex might support greater efficiency of print-to-sound conversion. Relationships were most apparent with phonological decoding ability, suggesting that the efficiency of nonword reading might relate to connectivity between temporal cortex and other regions, while sight word reading might rely less strongly on such interconnectivity. Dissociations were noted in how phonological decoding and comprehension measures predicted white matter tracts, suggesting that different aspects of reading relate to white matter anatomy in different regions. This dissociation mirrors functional imaging data that suggest that different cortical networks are engaged by different reading tasks and predict different reading skills. Overall, our study supports the view that atypical white matter organization associated with reading impairment may reflect the tail end of the distribution, as relationships between temporal white matter anatomy and reading ability were apparent across the spectrum of reading ability.

Supplementary data to this article can be found online at <http://dx.doi.org/10.1016/j.neuroimage.2014.03.069>.

References

- Beaulieu, C., Plewes, C., Paulson, L.A., Roy, D., Snook, L., Concha, L., Phillips, L., 2005. Imaging brain connectivity in children with diverse reading ability. *NeuroImage* 25 (4), 1266–1271.
- Behrens, T.E.J., Woolrich, M.W., Jenkinson, M., Johansen-Berg, H., Nunes, R.G., Clare, S., Smith, S.M., 2003a. Characterization and propagation of uncertainty in diffusion-weighted MR imaging. *Magn. Reson. Med.* 50 (5), 1077–1088.
- Behrens, T.E.J., Johansen-Berg, H., Woolrich, M.W., Smith, S.M., Wheeler-Kingshott, C.A.M., Boulby, P.A., Matthews, P.M., 2003b. Non-invasive mapping of connections between human thalamus and cortex using diffusion imaging. *Nat. Neurosci.* 6 (7), 750–757.
- Behrens, T.E.J., Berg, H.J., Jbabdi, S., Rushworth, M.F.S., Woolrich, M.W., 2007. Probabilistic diffusion tractography with multiple fibre orientations: what can we gain? *NeuroImage* 34 (1), 144–155.
- Blomert, L., Froyen, D., 2010. Multi-sensory learning and learning to read. *Int. J. Psychophysiol.* 77 (3), 195–204.
- Brambati, S.M., Termine, C., Ruffino, M., Stella, G., Fazio, F., Cappa, S.F., Perani, D., 2004. Regional reductions of gray matter volume in familial dyslexia. *Neurology* 63 (4), 742–745.
- Brown, J.I., Fisch, V.V., Hanna, G., 1993. *Nelson–Denny Reading Test*. Riverside, Chicago, IL.
- Brown, L., Sherbenou, R., Johnsen, S., 1997. *Test of Nonverbal Intelligence: A Language-Free Measure of Cognitive Ability*. Pro-ed, Austin, Texas.
- Brown, W.E., Eliez, S., Menon, V., Rumsey, J.M., White, C.D., Reiss, A.L., 2001. Preliminary evidence of widespread morphological variations of the brain in dyslexia. *Neurology* 56 (6), 781–783.
- Carter, J.C., Lanham, D.C., Cutting, L.E., Clements-Stephens, A.M., Chen, X., Hadzipasic, M., Kaufmann, W.E., 2009. A dual DTI approach to analyzing white matter in children with dyslexia. *Psychiatry Res. Neuroimaging* 172 (3), 215–219.
- Catani, M., Mesulam, M., 2008. The arcuate fasciculus and the disconnection theme in language and aphasia: history and current state. *Cortex* 44 (8), 953–961.
- Catani, M., Jones, D.K., Ffytche, 2005. Perisylvian language networks of the human brain. *Ann. Neurol.* 57 (1), 8–16.
- Chiang, M.C., Barysheva, M., Shattuck, D.W., Lee, A.D., Madsen, S.K., Avedissian, C., Thompson, P.M., 2009. Genetics of brain fiber architecture and intellectual performance. *J. Neurosci.* 29 (7), 2212–2224.
- Clements-Stephens, A.M., Materek, A.D., Eason, S.H., Scarborough, H.S., Pugh, K.R., Rimrodt, S., Cutting, L.E., 2012. Neural circuitry associated with two different approaches to novel word learning. *Dev. Cogn. Neurosci.* 2, S99–S113.
- Cohen, L., Dehaene, S., 2004. Specialization within the ventral stream: the case for the visual word form area. *NeuroImage* 22 (1), 466–476.
- Conturo, T.E., Lori, N.F., Cull, T.S., Akbudak, E., Snyder, A.Z., Shimony, J.S., Raichle, M.E., 1999. Tracking neuronal fiber pathways in the living human brain. *Proc. Natl. Acad. Sci.* 96 (18), 10422–10427.
- Dejerine, J.J., 1892. Contribution à l'étude anatomo-pathologique et clinique des différentes variétés de cécité verbale.
- Deutsch, G.K., Dougherty, R.F., Bammer, R., Siok, W.T., Gabrieli, J.D., Wandell, B., 2005. Children's reading performance is correlated with white matter structure measured by diffusion tensor imaging. *Cortex* 41 (3), 354–363.
- Dougherty, R.F., Ben-Shachar, M., Deutsch, G.K., Hernandez, A., Fox, G.R., Wandell, B.A., 2007. Temporal-callosal pathway diffusivity predicts phonological skills in children. *Proc. Natl. Acad. Sci.* 104 (20), 8556–8561.
- Dudink, J., Kerr, J.L., Paterson, K., Counsell, S.J., 2008. Connecting the developing preterm brain. *Early Hum. Dev.* 84 (12), 777–782.
- Duffau, H., Capelle, L., Sichez, N., Denvil, D., Lopes, M., Sichez, J.P., Fohanno, D., 2002. Intra-operative mapping of the subcortical language pathways using direct stimulations: An anatomo-functional study. *Brain* 125 (1), 199–214.
- Duffau, H., Gatignol, P., Mandonnet, E., Peruzzi, P., Tzourio-Mazoyer, N., Capelle, L., 2005. New insights into the anatomo-functional connectivity of the semantic system: a study using cortico-subcortical electrostimulations. *Brain* 128 (4), 797–810.
- Eckert, M.A., Leonard, C.M., Wilke, M., Eckert, M., Richards, T., Richards, A., Berninger, V., 2005. Anatomical signatures of dyslexia in children: unique information from manual and voxel based morphometry brain measures. *Cortex* 41 (3), 304–315.
- Eliez, S., Rumsey, J.M., Giedd, J.N., Schmitt, J.E., Patwardhan, A.J., Reiss, A.L., 2000. Morphological alteration of temporal lobe gray matter in dyslexia: an MRI study. *J. Child Psychol. Psychiatry* 41 (5), 637–644.
- Epelbaum, S., Pinel, P., Gaillard, R., Delmaire, C., Perrin, M., Dupont, S., Cohen, L., 2008. Pure alexia as a disconnection syndrome: new diffusion imaging evidence for an old concept. *Cortex* 44 (8), 962–974.
- Frye, R.E., Hasan, K., Xue, L., Strickland, D., Malmberg, B., Liederman, J., Papanicolaou, A., 2008. Splanium microstructure is related to two dimensions of reading skill. *Neuroreport* 19 (16), 1627.
- Gough, P.M., Nobre, A.C., Devlin, J.T., 2005. Dissociating linguistic processes in the left inferior frontal cortex with transcranial magnetic stimulation. *J. Neurosci.* 25 (35), 8010–8016.
- Grossman, M., McMillan, C., Moore, P., Ding, L., Glosser, G., Work, M., Gee, J., 2004. What's in a name: voxel-based morphometric analyses of MRI and naming difficulty in Alzheimer's disease, frontotemporal dementia and corticobasal degeneration. *Brain* 127 (3), 628–649.
- Hoef, F., McCandliss, B.D., Black, J.M., Gantman, A., Zakerani, N., Hulme, C., Gabrieli, J.D., 2011. Neural systems predicting long-term outcome in dyslexia. *Proc. Natl. Acad. Sci.* 108 (1), 361–366.
- Jenkinson, M., Smith, S., 2001. A global optimisation method for robust affine registration of brain images. *Med. Image Anal.* 5 (2), 143–156.
- Jobard, G., Crivello, F., Tzourio-Mazoyer, N., 2003. Evaluation of the dual route theory of reading: a meta-analysis of 35 neuroimaging studies. *NeuroImage* 20 (2), 693–712.
- Katzir, T., Misra, M., Poldrack, R.A., 2005. Imaging phonology without print: assessing the neural correlates of phonemic awareness using fMRI. *NeuroImage* 27 (1), 106–115.
- Keller, T.A., Just, M.A., 2009. Altering cortical connectivity: remediation-induced changes in the white matter of poor readers. *Neuron* 64, 624–631.
- Klingberg, T., Hedehus, M., Temple, E., Salz, T., Gabrieli, J.D., Moseley, M.E., Poldrack, R.A., 2000. Microstructure of temporo-parietal white matter as a basis for reading ability: evidence from diffusion tensor magnetic resonance imaging. *Neuron* 25 (2), 493–500.
- Lu, L.H., Crosson, B., Nadeau, S.E., Heilman, K.M., Gonzalez-Rothi, L.J., Raymer, A., Roper, S. N., 2002. Category-specific naming deficits for objects and actions: semantic attribute and grammatical role hypotheses. *Neuropsychologia* 40 (9), 1608–1621.
- Mandonnet, E., Nouet, A., Gatignol, P., Capelle, L., Duffau, H., 2007. Does the left inferior longitudinal fasciculus play a role in language? A brain stimulation study. *Brain* 130 (3), 623–629.
- McDermott, K.B., Petersen, S.E., Watson, J.M., Ojemann, J.G., 2003. A procedure for identifying regions preferentially activated by attention to semantic and phonological relations using functional magnetic resonance imaging. *Neuropsychologia* 41 (3), 293–303.
- Mori, S., Wakanas, S., Nagae-Poetscher, L.M., van Zijl, P.C., 2005. *MRI Atlas of Human White Matter*. Elsevier, Amsterdam.
- Moss, J., Schunn, C.D., Schneider, W., McNamara, D.S., VanLehn, K., 2011. The neural correlates of strategic reading comprehension: cognitive control and discourse comprehension. *NeuroImage* 58 (2), 675–686.
- Niogi, S.N., McCandliss, B.D., 2006. Left lateralized white matter microstructure accounts for individual differences in reading ability and disability. *Neuropsychologia* 44 (11), 2178–2188.
- Odegard, T.N., Farris, E.A., Ring, J., McColl, R., Black, J., 2009. Brain connectivity in non-reading impaired children and children diagnosed with developmental dyslexia. *Neuropsychologia* 47 (8), 1972–1977.
- Pannek, K., Mathias, J.L., Bigler, E.D., Brown, G., Taylor, J.D., Rose, S., 2010. An automated strategy for the delineation and parcellation of commissural pathways suitable for clinical populations utilising high angular resolution diffusion imaging tractography. *NeuroImage* 50 (3), 1044–1053.
- Park, H.J., Kim, J.J., Lee, S.K., Seok, J.H., Chun, J., Kim, D.I., Lee, J.D., 2008. Corpus callosal connection mapping using cortical gray matter parcellation and DT-MRI. *Hum. Brain Mapp.* 29 (5), 503–516.
- Price, C.J., Moore, C.J., Humphreys, G.W., Wise, R.J.S., 1997. Segregating semantic from phonological processes during reading. *J. Cogn. Neurosci.* 9 (6), 727–733.
- Pugh, K.R., Shaywitz, B.A., Shaywitz, S.E., Constable, R.T., Skudlarski, P., Fulbright, R.K., Gore, J.C., 1996. Cerebral organization of component processes in reading. *Brain* 119 (4), 1221–1238.
- Richards, T., Stevenson, J., Crouch, J., Johnson, L.C., Maravilla, K., Stock, P., Berninger, V., 2008. Tract-based spatial statistics of diffusion tensor imaging in adults with dyslexia. *Am. J. Neuroradiol.* 29 (6), 1134–1139.
- Sandak, R., Frost, S.J., Rueckl, J.G., Landi, N., Mencl, W.E., Katz, L., Pugh, K.R., 2012. How does the brain read words? In: Spivey, M., McRae, K., Joanisse, M. (Eds.), *The Cambridge Handbook of Psycholinguistics*. Cambridge University Press, New York, NY, pp. 218–237.
- Saur, D., Kreher, B.W., Schnell, S., Kümmerer, D., Kellmeyer, P., Vry, M.S., Weiller, C., 2008. Ventral and dorsal pathways for language. *Proc. Natl. Acad. Sci.* 105 (46), 18035–18040.
- Schmithorst, V.J., Wilke, M., Dardzinski, B.J., Holland, S.K., 2005. Cognitive functions correlate with white matter architecture in a normal pediatric population: a diffusion tensor MRI study. *Hum. Brain Mapp.* 26 (2), 139–147.

- Schwartzman, A., Dougherty, R.F., Taylor, J.E., 2005. Cross-subject comparison of principal diffusion direction maps. *Magn. Reson. Med.* 53 (6), 1423–1431.
- Smith, S.M., 2002. Fast robust automated brain extraction. *Hum. Brain Mapp.* 17 (3), 143–155.
- Smith, S.M., Jenkinson, M., Woolrich, M.W., Beckmann, C.F., Behrens, T.E., Johansen-Berg, H., Matthews, P.M., 2004. Advances in functional and structural MR image analysis and implementation as FSL. *NeuroImage* 23, S208–S219.
- Steinbrink, C., Vogt, K., Kastrup, A., Müller, H.P., Juengling, F.D., Kassubek, J., Riecker, A., 2008. The contribution of white and gray matter differences to developmental dyslexia: insights from DTI and VBM at 3.0 T. *Neuropsychologia* 46 (13), 3170–3178.
- Tang, C.Y., Eaves, E.L., Ng, J.C., Carpenter, D.M., Mai, X., Schroeder, D.H., Haier, R.J., 2010. Brain networks for working memory and factors of intelligence assessed in males and females with fMRI and DTI. *Intelligence* 38 (3), 293–303.
- Torgesen, J.K., Wagner, R.K., Rashotte, C.A., 1999. *Test of Word Reading Efficiency*. Pro-ed, Austin, Texas.
- van Atteveldt, N., Formisano, E., Goebel, R., Blomert, L., 2004. Integration of letters and speech sounds in the human brain. *Neuron* 43 (2), 271–282.
- Vandermosten, M., Boets, B., Poelmans, H., Snaert, S., Wouters, J., Ghesquière, P., 2012. A tractography study in dyslexia: neuroanatomic correlates of orthographic, phonological and speech processing. *Brain* 135 (3), 935–948.
- Vinckenbosch, E., Robichon, F., Eliez, S., 2005. Gray matter alteration in dyslexia: converging evidence from volumetric and voxel-by-voxel MRI analyses. *Neuropsychologia* 43 (3), 324–331.
- Welcome, S.E., Joanisse, M.F., 2012. Individual differences in skilled adult readers reveal dissociable patterns of neural activity associated with component processes of reading. *Brain Lang.* 120 (3), 360–371.
- Welcome, S.E., Chiarello, C., Thompson, P.M., Sowell, E.R., 2011. Reading skill is related to individual differences in brain structure in college students. *Hum. Brain Mapp.* 32 (8), 1194–1205.
- Wernicke, C., 1874. *Der aphasische Symptomenkomplex*. M. Cohn und Weigart, Breslau.
- Westerhausen, R., Grüner, R., Specht, K., Hugdahl, K., 2009. Functional relevance of inter-individual differences in temporal lobe callosal pathways: a DTI tractography study. *Cereb. Cortex* 19 (6), 1322–1329.
- Witelson, S.F., 1989. Hand and sex differences in the isthmus and genu of the human corpus callosum: a postmortem morphological study. *Brain* 112 (3), 799–835.
- Yeatman, J.D., Dougherty, R.F., Rykhlevskaia, E., Sherbondy, A.J., Deutsch, G.K., Wandell, B.A., Ben-Shachar, M., 2011. Anatomical properties of the arcuate fasciculus predict phonological and reading skills in children. *J. Cogn. Neurosci.* 23 (11), 3304–3317.
- Yeatman, J.D., Dougherty, R.F., Ben-Shachar, M., Wandell, B.A., 2012. Development of white matter and reading skills. *Proc. Natl. Acad. Sci.* 109 (44), E3045–E3053.
- Zatorre, R.J., Meyer, E., Gjedde, A., Evans, A.C., 1996. PET studies of phonetic processing of speech: review, replication, and reanalysis. *Cereb. Cortex* 6 (1), 21–30.
- Zhang, Y., Brady, M., Smith, S., 2001. Segmentation of brain MR images through a hidden Markov random field model and the expectation-maximization algorithm. *IEEE Trans. Med. Imaging* 20 (1), 45–57.